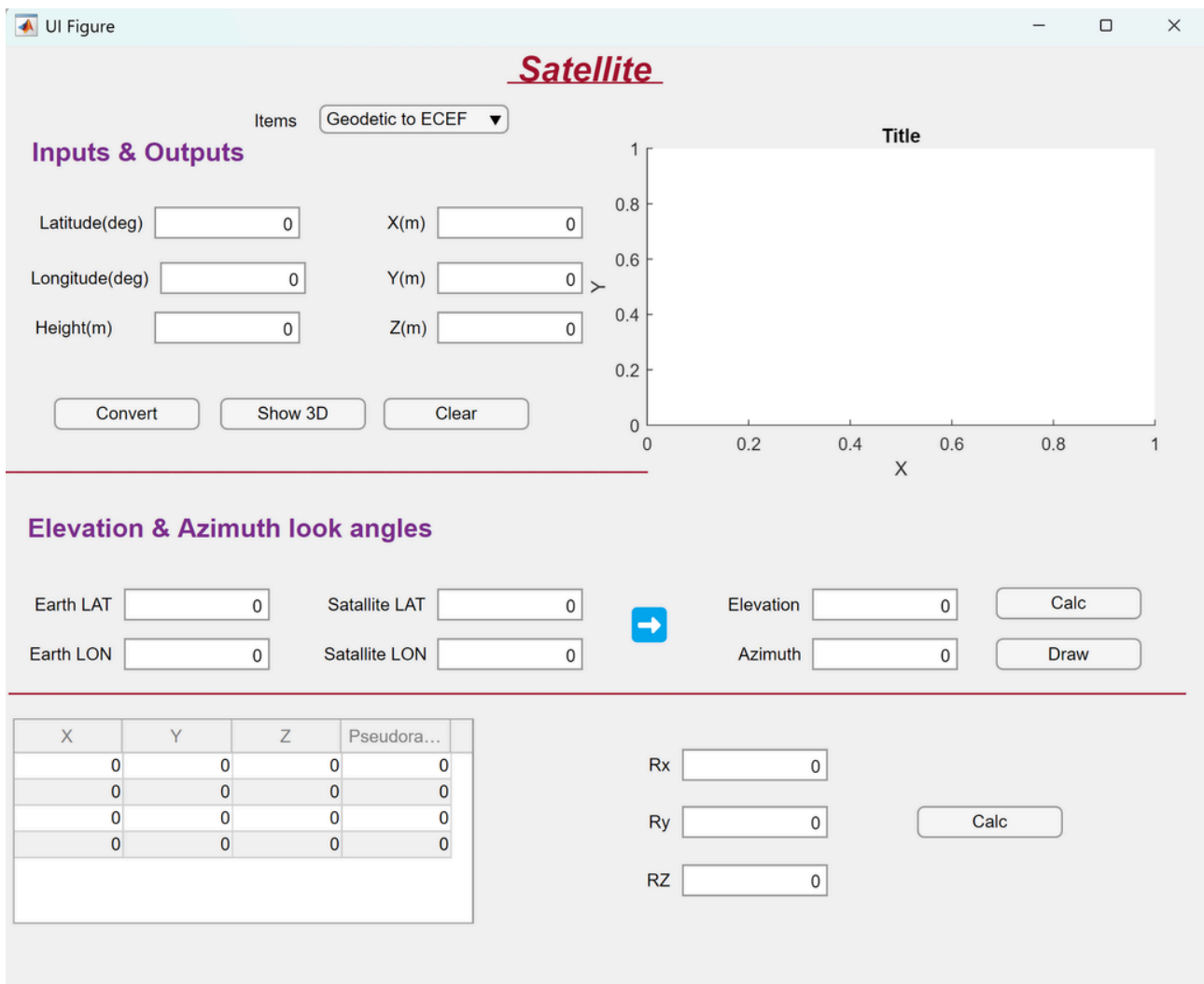


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OBJECTIVE

The aim of this project is to design a MATLAB GUI application that performs:

- Conversion between Geodetic coordinates and ECEF coordinates
- Calculation of Elevation and Azimuth angles
- Determination of GPS receiver position using pseudorange data
- Visualization of positions in a 3D Earth model

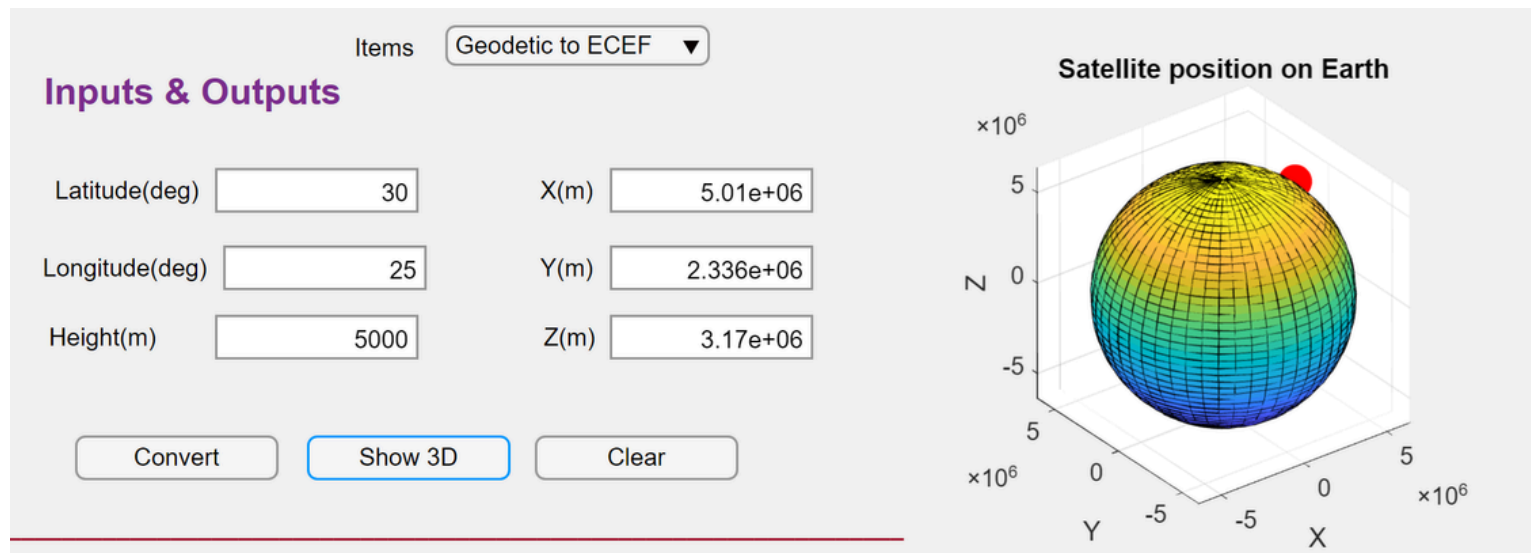
TOOLS USED

- MATLAB
- App Designer (GUI)
- Mathematical modeling of Earth (WGS84)

SYSTEM COMPONENTS

The GUI consists of three main parts:

PART 1: GEODETIC ↔ ECEF CONVERSION



This part converts between:

- Latitude (φ), Longitude (λ), Height (h)
- Cartesian coordinates (X , Y , Z)

Equations Used

$$N = \frac{a}{\sqrt{1 - e^2 \sin^2 \phi}}$$

$$X = (N + h) \cos \phi \cos \lambda$$

$$Y = (N + h) \cos \phi \sin \lambda$$

$$Z = (N(1 - e^2) + h) \sin \phi$$

$$a = 6378137 \text{ m}$$

$$b = 6356752 \text{ m}$$

$$e^2 = 2f - f^2$$

$$f = \frac{a-b}{a}$$

GUI Features

- User inputs Latitude, Longitude, Height
- Output: X , Y , Z
- Option to reverse conversion

PART 2: ELEVATION & AZIMUTH ANGLES

Elevation & Azimuth look angles

Earth LAT

Satellite LAT

Earth LON

Satellite LON



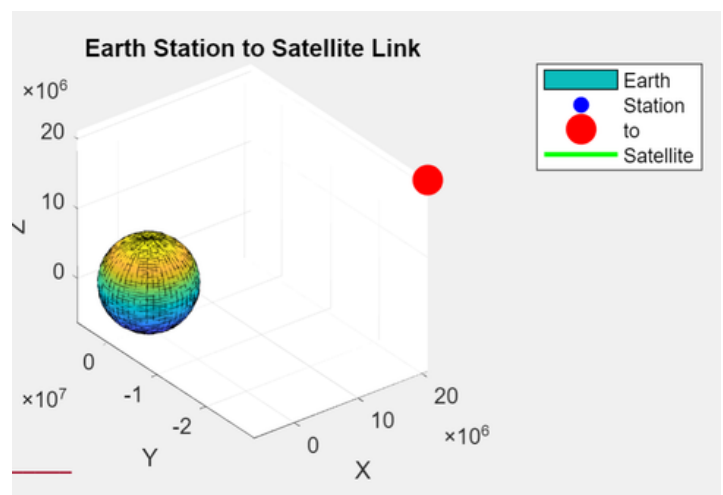
Elevation

Calc

Azimuth

Draw

Calculates satellite look angles from an Earth station.



Inputs

- Earth station (Lat, Lon)
- Satellite (Lat, Lon)

Equations

- Central Angle: $\cos(\gamma) = \cos(L_e) \cos(L_s) \cos(l_s - l_e) + \sin(L_e) \sin(L_s)$
- Elevation: $\cos(E) = \frac{\sin(\gamma)}{\sqrt{1 + (r_e/r_s)^2 - 2(r_e/r_s) \cos(\gamma)}}$
- Azimuth: $\alpha = \tan^{-1} \left(\frac{\tan(l_s - l_e)}{\sin(L_e)} \right)$
 $Az = 180^\circ + \alpha$

Output

- Elevation angle
- Azimuth angle

PART 3: GPS POSITION CALCULATION

X	Y	Z	Pseudora...
30	20	0	50
20	40	0	30
80	20	0	20
50	60	0	100

Rx

Ry

RZ

Global Positioning System (GPS) is a satellite-based navigation system that allows users to determine their exact location anywhere on Earth. The system works by receiving signals from at least four satellites. Each satellite provides its position and the distance (pseudorange) to the user. By combining these measurements, we can calculate the user's position in 3D space.

Equations

- $A = 2 * \begin{bmatrix} (x_4 - x_1) & (y_4 - y_1) & (z_4 - z_1) \\ (x_4 - x_2) & (y_4 - y_2) & (z_4 - z_2) \\ (x_4 - x_3) & (y_4 - y_3) & (z_4 - z_3) \end{bmatrix}$
- $B = \begin{bmatrix} (r_1^2 - r_4^2) - (x_1^2 - x_4^2) - (y_1^2 - y_4^2) - (z_1^2 - z_4^2) \\ (r_2^2 - r_4^2) - (x_2^2 - x_4^2) - (y_2^2 - y_4^2) - (z_2^2 - z_4^2) \\ (r_3^2 - r_4^2) - (x_3^2 - x_4^2) - (y_3^2 - y_4^2) - (z_3^2 - z_4^2) \end{bmatrix}$
- $X = \text{pinv}(A) * B$

CONCLUSION

The project demonstrates integration of:

- Mathematical modeling
- Satellite communication concepts
- Numerical methods (Least Squares)
- GUI design

This system simulates real-world GPS positioning and satellite tracking.